

ECO101H1F L101, L201, L301 & L501: Principles of Microeconomics
Midterm 1, Fall 2025

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Multiple Choice Questions

PUT MULTIPLE CHOICE ANSWERS IN THE BUBBLE SHEET (LAST PAGE).

No work here will be considered.

- 1) [2 points] Aloysius purchased a \$15 non-refundable ticket for a 3-hour movie he values at \$50. If he doesn't go to the movie, he can spend the 3 hours with his grandchildren (valued at \$45) or watching the Blue Jays baseball game on TV (valued at \$20). His Opportunity Cost of going to the movie is ____.
- A. \$15
B. \$20
C. \$45
D. \$65
E. None of the above.

Solution: C. He has 2 alternatives, both using the same resource of 3 hours. Opportunity Cost is the value of the next best forgone alternative use of that time: \$45 (grandchildren). The \$15 is sunk and irrelevant.

- 2) [2 points] Janice and Joe's Marginal Cost (MC) for the Q th unit is given in the table below. From the options below, at what price will there be a total of exactly 5 units supplied?

Q	1	2	3	4	5
Janice's MCs	\$0	\$2	\$4	\$6	\$6
Joe's MCs	\$3	\$5	\$7	\$8	\$9

- A. \$7.5
B. \$6.5
C. \$5.5
D. \$3.5
E. None of the above.

Solution: C. Sort all unit MCs across both sellers: $\{0, 2, 3, 4, 5, 6, 6, 7, 8, 9\}$. The 5 lowest-MC units are 0, 2, 3, 4 and 5. Any price at or above \$5 and below \$6 induces exactly those 5 units to be supplied.

- 3) [2 points] The table below has the amount of time Earl and Pearl need to re-stock one shelf or clean one closet. Each has 100 hours of time. Pearl's Opportunity Cost of cleaning one closet is ____ restocked shelves.

	Re-stock one shelf	Clean one closet
Earl	4 hours	2 hours
Pearl	3 hours	4 hours

- A. 4/3 shelves
B. 3/4 of a shelf

- C. 2 shelves
- D. 1/2 a shelf
- E. None of the above.

Solution: A. Cleaning one closet takes Pearl 4 hours. In those same 4 hours she could instead have restocked $4/3$ shelves, so her Opportunity Cost of one closet is $4/3$ shelves.

- 4) [2 points] Omar and Priya produce coffee and cocoa on identical plots of land. Priya can produce more of both coffee and cocoa per hectare than Omar. Omar exports coffee to Priya in exchange for cocoa. Based on this, ____ has the Absolute Advantage in cocoa, and ____ has the Comparative Advantage in coffee.
- A. Omar; Omar
 - B. Priya; Omar
 - C. Omar; Priya
 - D. Priya; Priya
 - E. Cannot be determined; Omar

Solution: B. Producing more of both means Priya has the Absolute Advantage in both goods, including cocoa. The trade pattern reveals Comparative Advantage: the exporter has the Comparative Advantage in the good it exports, so Omar has the Comparative Advantage in coffee.

- 5) [2 points] Assume perfectly competitive markets where P is price and Q is quantity. Market Demand is $P = 220 - 2Q$ and Market Supply is $P = 20 + 2Q$. What is equilibrium market price?
- A. \$50
 - B. \$60
 - C. \$85
 - D. \$120
 - E. None of the above.

Solution: D. Set demand equal to supply: $220 - 2Q = 20 + 2Q \Rightarrow 200 = 4Q \Rightarrow Q^* = 50$. Substituting back, $P^* = 20 + 2(50) = \$120$.

- 6) [2 points] In a perfectly competitive market, we have 2 firms, A and B, whose Marginal Cost (MC) functions are $MC_A = 2Q_A + 20$ and $MC_B = Q_B + 10$, respectively. At a market price of \$40, market quantity supplied is ____.
- A. 20
 - B. 40
 - C. 60
 - D. 90
 - E. None of the above.

Solution: B. Each firm supplies where price equals its MC. Firm A: $2Q_A + 20 = 40 \Rightarrow Q_A = 10$. Firm B: $Q_B + 10 = 40 \Rightarrow Q_B = 30$. Total $Q = 10 + 30 = 40$.

- 7) [2 points] Consider the perfectly competitive green tea market. Demand is downward-sloping. The elasticity of supply is 0. A study showing the multiple health benefits of drinking green tea goes viral on social media. As a result, we would expect market price for green tea to ____ and market quantity to ____.
- A. stay the same; decrease.
 - B. decrease; decrease.

- C. increase; increase.
- D. stay the same; increase.
- E. increase; stay the same.

Solution: E. An elasticity of supply of 0 means supply is vertical. The viral study raises MWTP, so demand shifts out. With vertical supply, a demand shift raises price but leaves quantity unchanged.

8) [2 points] Market demand is given by $P = 60 - 3Q$. At a price of $P = \$15$, market demand is ____.

- A. Perfectly inelastic
- B. Inelastic
- C. Unit elastic
- D. Elastic
- E. Perfectly elastic

Solution: B. At $P = \$15$, $15 = 60 - 3Q \Rightarrow Q = 15$. From $P = 60 - 3Q$ we get $\left| \frac{\Delta Q}{\Delta P} \right| = \frac{1}{3}$, so

$$\varepsilon_D = \left| \frac{\Delta Q}{\Delta P} \right| \frac{P}{Q} = \frac{1}{3} \cdot \frac{15}{15} = 0.33 < 1,$$

which is inelastic.

9) [2 points] Consider the perfectly competitive coffee market with downward-sloping demand and upward-sloping supply. Consumers choose between coffee and tea. The price of tea decreases, all else held fixed. As a result, in the coffee market Total Surplus ____ and Consumer Surplus ____.

- A. decreases; decreases.
- B. increases; increases.
- C. decreases; increases.
- D. increases; could increase, decrease or stay the same.
- E. decreases; could increase, decrease or stay the same.

Solution: E. A cheaper substitute lowers MWTP for coffee, which shifts coffee demand in/down, lowering both P and Q . Lower MWTP and lower Q mean Total Surplus falls. For Consumer Surplus, MWTP falls (pushing CS down) but P also falls (pushing CS up), so the net effect is uncertain.

10) [2 points] Assume buyers only buy 1 unit, if they buy, and sellers sell only 1 unit, if they sell. The table below has the Marginal Costs (MC) for Producers A, B and C and the Willingness to Pay (WTP) for buyers X, Y and Z. If they are the only participants in a perfectly competitive market, Total Surplus is ____.

Buyer	WTP	Seller	MC
X	\$9	A	\$13
Y	\$20	B	\$9
Z	\$11	C	\$15

- A. \$7
- B. \$9
- C. \$11
- D. \$13
- E. None of the above.

Solution: C. Arrange WTP from highest to lowest (\$20, \$11, \$9) and MC from lowest to highest (\$9, \$13, \$15). The first unit trades: $\$20 - \$9 = \$11$ of surplus. The second unit would not, since $\$11 < \13 . So 1 unit trades and Total Surplus is \$11.

- 11) [3 points] Adele, Belle, Darnell and Marcel's Willingness to Pay (WTP) for one concert ticket is given below. Tickets are priced at \$180 each. If the ticket price increases to \$200, all else the same, Total Surplus _____ and Consumer Surplus _____.

Person	Adele	Belle	Darnell	Marcel
WTP	\$400	\$150	\$100	\$250

- A. decreases; decreases by \$40.
- B. decreases; increases by \$40.
- C. increases; increases by \$40.
- D. stays the same; decreases by \$40.
- E. stays the same; increases by \$40.

Solution: D. At \$180 the buyers are Adele and Marcel. At \$200 both still buy ($WTP \geq \200), so quantity is unchanged and Total Surplus is unchanged: a price change is a transfer between buyers and sellers. Consumer Surplus falls by \$40, since each of the two buyers loses \$20: from $(400 - 180) + (250 - 180) = \290 to $(400 - 200) + (250 - 200) = \250 .

- 12) [3 points] Homer bought a concert ticket for \$250. His value from attending the concert is \$400. If he returns the ticket, Homer can get a refund of \$100. The day of the concert, Ned offers Homer \$70 for his ticket, while Moe offers him \$120. Homer's sunk cost of the concert ticket is...
- A. \$100
 - B. \$120
 - C. \$150
 - D. \$400
 - E. None of the above.

Solution: E. The sunk cost is the part of the \$250 that cannot be recovered. He can recover at most $\max\{\$100, \$70, \$120\} = \120 , so the sunk cost is $\$250 - \$120 = \$130$, which is not one of the listed options.

- 13) [3 points] Two producers, Aria and Ben. Their labor requirements to produce one gadget or one widget are below. Based on this information, which exchange rate makes both Aria and Ben better off?

Person	One Gadget	One Widget
Aria	10 hours	5 hours
Ben	12 hours	4 hours

- A. 10 gadgets for 5 widgets
- B. 10 gadgets for 15 widgets
- C. 10 gadgets for 25 widgets
- D. 10 gadgets for 30 widgets
- E. 10 gadgets for 35 widgets

Solution: C. In one hour each can produce:

Person	Gadgets	Widgets
Aria	1/10	1/5
Ben	1/12	1/4

This makes the Opportunity Cost of 10 gadgets:

Person	OC of 10 Gadgets
Aria	20 widgets
Ben	30 widgets

Aria has the Comparative Advantage in gadgets, so she exports them. A trade that makes both better off must lie strictly between 20 and 30 widgets for 10 gadgets. Only 25 widgets qualifies.

- 14) [3 points] Aria and Ben have 100 hours of time each to produce either widgets or gadgets. The amount of time required to produce one gadget or one widget is given below. Working together they are currently producing 12 gadgets. Assuming efficient production, if they produced an extra gadget, ____ would make it, at an Opportunity Cost of ____ widgets.

Person	One Gadget	One Widget
Aria	10 hours	5 hours
Ben	12 hours	4 hours

- A. Ben; 3 widgets
- B. Aria; 3 widgets
- C. Ben; 0.33 widgets
- D. Aria; 0.5 widgets
- E. None of the above.

Solution: A. Aria's Opportunity Cost of a gadget is 2 widgets and Ben's is 3 widgets, so efficient production uses Aria's time first. With 100 hours she can make at most $100/10 = 10$ gadgets. Producing 12 gadgets efficiently therefore means Aria makes 10 (all her time) and Ben makes 2. The next gadget must be made by Ben, costing him 12 hours that could instead have produced $12/4 = 3$ widgets.

- 15) [3 points] In the perfectly competitive food delivery market, consumers who do not want to eat in a restaurant can hire firms to deliver meals to their home. They pay \$5 per meal as a delivery fee. In Winter 2026, two things happen: (1) demand for meals delivered falls and (2) at every delivery fee, the number of delivery firms in the market increases. As a result, we expect the market delivery fee to ____ and market quantity to ____.

- A. decrease; stay the same.
- B. could increase, decrease or stay the same; decrease.
- C. could increase, decrease or stay the same; increase.
- D. increase; could increase, decrease or stay the same.
- E. decrease; could increase, decrease or stay the same.

Solution: E. Demand shifts in and supply shifts out. Both changes push the delivery fee down, so price falls. Demand shifting in lowers quantity while supply shifting out raises it, so quantity is uncertain.

- 16) [3 points] Assume perfectly competitive markets with downward-sloping demand and upward-sloping supply curves. Consumers choose between cereal and smoothies for breakfast. Wheat is used in cereal production, not smoothie production. When the price of wheat rises, all else held fixed, the price of cereal ____ and the price of smoothies ____.

- A. rises; rises.
- B. falls; rises.
- C. rises; falls.
- D. falls; falls.
- E. rises; remains unchanged.

Solution: A. An increase in MC in the cereal market shifts cereal supply up/in and raises the price of cereal. A higher price for the substitute raises MWTP in the smoothie market, so smoothie demand shifts out and the price of smoothies also rises.

- 17) [3 points] Assume competitive markets with typical curves. Data shows that when the price of mobile phones rises, all else held fixed, the quantity demanded of smartwatches falls. This means that the cross-price elasticity of smartwatches and phones is ____ as consumers consider them ____.
- A. positive; substitutes
 - B. positive; complements
 - C. negative; substitutes
 - D. negative; complements
 - E. negative; inferior

Solution: D. Price of phones $\uparrow \Rightarrow$ demand for smartwatches $\downarrow \Rightarrow$ negative cross-price elasticity $\Rightarrow$ the goods are complements.

- 18) [3 points] Assume no externalities. We have 4 potential buyers whose Marginal Willingness to Pay (MWTP) for one ticket is given in the table below. Only two tickets, priced at \$200 each, are available and A and B buy them. If A sells their ticket to D for \$450, Total Surplus ____.

Buyer	A	B	C	D
MWTP	\$450	\$250	\$350	\$500

- A. Increases by \$50
- B. Increases by \$200
- C. Increases by \$250
- D. Decreases by \$200
- E. None of the above.

Solution: A. With fixed supply and no production costs, Total Surplus equals the sum of the final holders' valuations; the prices paid are transfers. Initially the holders are A and B: $450 + 250 = \$700$. After the resale the holders are D and B: $500 + 250 = \$750$. The change is $750 - 700 = +\$50$.

- 19) [3 points] Suppose that when average family income rises from \$60,000 per year to \$80,000 per year, all else the same, average family annual spending on internet services rises from \$1,200 to \$1,500. Based on this information, over this change, internet service is ____ and the own-price elasticity of demand for internet services ____.
- A. a normal good; inelastic
 - B. a normal good; cannot be determined.
 - C. an inferior good; cannot be determined
 - D. a normal good; elastic
 - E. an inferior good; elastic

Solution: B. A change in income, all else held fixed, shifts the demand curve and moves us along the supply curve, so price and quantity move in the same direction. Spending is $P \times Q$, and spending rose, so both price and quantity must have risen. That is a movement up the supply curve, which means demand shifted out when income rose: internet service is a normal good. Because this is a shift of demand rather than a movement along it, the data tell us nothing about the own-price elasticity of demand.

- 20) [3 points] Fatimah's Total Benefits from visiting the doctor throughout the year are given below. She is given the choice between paying \$200 per visit and paying \$1,000 for the entire year with unlimited visits. How many times will Fatimah visit the doctor?

Number of visits	1	2	3	4	5
Total Benefit	\$500	\$825	\$1,000	\$1,150	\$1,150

- A. 5
- B. 4
- C. 3
- D. 2
- E. 1

Solution: D. The Marginal Benefits implied by the table are:

Number of visits	1	2	3	4	5
Marginal Benefit	\$500	\$325	\$175	\$150	\$0

Under the \$1,000 unlimited plan the MC of a visit is zero, so she visits 5 times for a net benefit of $\$1,150 - \$1,000 = \$150$. Paying \$200 per visit, she visits whenever $MB \geq \$200$, which is twice, for a net benefit of $\$825 - \$400 = \$425$. Since $\$425 > \150 , she chooses to pay per visit and goes 2 times.

PUT MULTIPLE CHOICE ANSWERS IN THE BUBBLE SHEET (LAST PAGE).

No work here will be considered.

Short Answer Questions

- Q1) Gus chooses between a sandwich from Urban Crust (priced at \$20) and a salad (priced at \$12). Both need a 15-minute walk to reach. Only Urban Crust has an additional 30-minute wait to get in.
- Q1a) [5 points] **Claim:** When Urban Crust's sandwich price falls from \$20 to \$15, Gus's Willingness to Pay (WTP) for the sandwich increases. Agree, disagree, or it depends? Use an equation for WTP to justify your answer.

Solution: Disagree. Gus decides to buy the sandwich by comparing the benefits and the Opportunity Cost of the sandwich:

$$\$V_{\text{Sandwich}} \text{ vs. } \$OC_{\text{Sandwich}}$$

Expanding the Opportunity Cost gives

$$\$V_{\text{Sandwich}} \text{ vs. } \$P_{\text{Sandwich}} + 45\text{mins} + \$V_{\text{Salad}} - \$P_{\text{Salad}} - 15\text{mins}$$

Rearranging so that everything other than the sandwich price is on the left:

$$\$V_{\text{Sandwich}} - \$V_{\text{Salad}} + \$P_{\text{Salad}} - 30\text{mins} \text{ vs. } \$P_{\text{Sandwich}}$$

This means

$$WTP_{\text{Sandwich}} = \$V_{\text{Sandwich}} - \$V_{\text{Salad}} + \$P_{\text{Salad}} - 30\text{mins}$$

From this we can see that the price of the sandwich does not appear in his WTP for the sandwich, so a change in that price does not change his WTP. He compares his WTP to the price to determine whether he should buy the sandwich. When the price changes, his decision may change, but that is because the price side of the comparison has changed, not the WTP side.

Q1b) [6 points] Gus will buy the Urban Crust sandwich. He now chooses between going to the restaurant or getting the same \$20 sandwich, delivered. The delivery fee is \$10. What is the maximum delivery fee Gus will pay? Explain.

Solution: Intuitively, the benefit of delivery is saving 45 mins of time, so he should be willing to pay up to the value of 45 mins of his time as a delivery fee. You would need information on the \$ value of 45 mins of his time to determine the actual number.

Using equations, he gets the sandwich delivered if

$$\$V_{delivery} \geq \$OC_{delivery}$$

He gets the same sandwich either way, so with $\$D$ as the delivery fee this becomes

$$\begin{aligned} \$V_{Sandwich} &\geq \$P_{Sandwich} + \$D + \$V_{Sandwich} - \$45mins - \$P_{Sandwich} \\ \Rightarrow \$V_{Sandwich} - \$V_{Sandwich} &\geq \$P_{Sandwich} - \$P_{Sandwich} - \$45mins + \$D \\ \$45mins &\geq \$D \end{aligned}$$

So the maximum delivery fee he will pay is the \$ value of 45 mins of his time. Since the actual fee is \$10, he takes delivery if 45 mins of his time is worth at least \$10.

Q2) Minnesota (MN) and British Columbia (BC) produce apples and corn with 1,000 acres each. Output per acre in 2025 is:

Region	Apples	Corn
Minnesota (MN)	7 tons per acre	21 tons per acre
British Columbia (BC)	8 tons per acre	16 tons per acre

Q2a) [6 points] MN offers to sell apples to BC at an exchange rate of 1 ton of apples for 3 tons of corn. **Claim:** Trading with MN at this rate will make BC better off. Agree, disagree, or it depends? Explain.

Solution: Disagree. The Opportunity Costs are:

	OC of 1 ton of Apples	OC of 1 ton of Corn
Minnesota (MN)	3 tons of corn	0.33 tons of apples
British Columbia (BC)	2 tons of corn	0.5 tons of apples

It costs BC 2 tons of corn to produce 1 ton of apples itself. The proposed trade asks for 3 tons of corn for 1 ton of apples, so it is cheaper for BC to produce the apples itself than to buy them from MN.

Equivalently, to produce the 3 tons of corn MN wants in exchange for 1 ton of apples, BC has to give up $3 \times \frac{1}{2} = 1.5$ tons of apples. It gets only 1 ton of apples in return, which is less than its own cost of production.

BC should reject MN's offer, as it makes them worse off.

Q2b) [5 points] Ignore part (a). New scenario. In 2026, good weather increases BC's corn productivity, all else the same. **Claim:** In 2026, BC should export corn to MN. Agree, disagree, or it depends? Explain.

Solution: It depends. Suppose BC's new productivity in corn is X tons per acre. The Opportunity Costs of corn are then:

	OC of 1 ton of Corn
Minnesota (MN)	$\frac{7}{21} = \frac{1}{3} = 0.33$ tons of apples
British Columbia (BC)	$\frac{8}{X}$ tons of apples

When X increases, BC's Opportunity Cost of corn falls. But that does not automatically mean BC now has the Comparative Advantage and starts exporting corn to MN. BC will only have the Comparative Advantage in corn if the productivity increase is large enough to push its OC below MN's:

$$\frac{1}{3} > \frac{8}{X} \iff X > 8 \times 3 = 24 \text{ tons per acre.}$$

So BC should export corn only if the good weather raises its corn yield above 24 tons per acre.

Q3) Assume all markets are perfectly competitive with the usual market demand and supply curves. Farmers raise chickens that lay eggs (sold in the egg market). They also sell some chickens as meat (chicken market). Bird flu kills a large fraction of chickens.

Q3a) [5 points] **Claim:** Prices in the egg market will fall due to the bird flu. Agree, disagree, or it depends? Explain your reasoning.

Solution: Disagree. The supply of eggs comes from chickens. If the chickens die then there are fewer eggs in the market at every market price; equivalently, the MC of an egg is higher at every quantity. This shifts the market supply curve up/in. The demand curve is unchanged, so we move up the demand curve to a higher price and a lower market quantity. At the old market price there would be excess demand for eggs, and price would have to rise to bring quantity demanded back in line with quantity supplied.

Q3b) [6 points] After the bird flu, spending on eggs rises while spending on chicken meat falls. **Claim:** This is because consumers have more alternatives for chicken meat than for eggs. Agree, disagree, or it depends? Explain.

Solution: Agree. When consumers have more alternatives their demand is more elastic, because they can switch away more easily when the price rises. This does not necessarily mean demand for chicken is elastic in levels; it could simply be more elastic than demand for eggs. Whether we can use the spending data depends on what is moving.

In both markets the bird flu shifts supply up/in: fewer surviving chickens means fewer chickens (and fewer eggs) brought to market at the same price, and the cost of an extra chicken or egg is higher at every quantity. Demand is unchanged in both markets, so we move up the demand curve to a higher price and a lower quantity. Because this is a movement along demand, we can use demand elasticity to sign the change in spending.

Spending is $P \times Q$. Spending on eggs rises when price rises, so demand for eggs is inelastic. Spending on chicken falls when price rises, so demand for chicken is elastic. Chicken demand being more elastic than egg demand is exactly what having more alternatives would produce, so the claim is consistent with the data.

Q3c) [6 points] 2029: bird flu is over and markets have returned to their pre-bird flu market equilibrium. Ann buys chicken and beef, no eggs. Holding all else constant, when her income rises by 10%, her quantity demanded of chicken increases by 12% and beef increases by 2%. **Claim:** Ann's demand for beef is elastic. Agree, disagree, or it depends? Explain.

Solution: Disagree. This question gives us the change in quantity demanded when income changes, all else held fixed. That is information on income elasticity:

$$\varepsilon_I = \frac{\% \Delta Q_D}{\% \Delta I}.$$

Own-price elasticity of demand instead has price changing, all else held fixed:

$$\varepsilon_D = \frac{\% \Delta Q_D}{\% \Delta P}.$$

We would need information on movements along the demand curve. We know both goods are normal, since quantity demanded rises with income. But because this is a shift of the demand curve rather than a movement along it, we cannot say anything about the own-price elasticity of demand for either beef or chicken.

Q4) FIFA must set ticket prices for World Cup 2026 matches. Assume: (1) all tickets are priced at \$100 each, (2) each match has 1,000 seats, and (3) at a price of \$100, many more than 1,000 tickets are demanded. Current plan: tickets are sold online, and the first 1,000 people to log in may buy one ticket each.

Q4a) [5 points] **Proposal A:** Everyone who wants a \$100 ticket signs up online. 1,000 names are randomly drawn and may purchase one ticket each. **Claim:** Producer Surplus will increase under Proposal A relative to the current plan. Agree, disagree, or it depends? Explain.

Solution: Disagree. Producer Surplus is the difference between price and MC for every unit sold. Under Proposal A the same 1,000 seats are sold, so quantity is unchanged, and the cost of producing those seats does not depend on who sits in them. They are all still sold at the same \$100 price. Producer Surplus is therefore unchanged.

Q4b) [6 points] Ignore part (a). **Proposal B:** Set the ticket price so that exactly 1,000 people are willing to buy a ticket at that price. **Claim:** Consumer Surplus will increase under Proposal B relative to the current plan. Agree, disagree, or it depends? Explain.

Solution: It depends. Consumer Surplus is the difference between MWTP and P for every unit sold. We have the same 1,000 seats, so quantity stays the same, but the price is no longer \$100: since quantity demanded at \$100 exceeds 1,000, the price must rise until $Q_D(P) = 1,000$. This higher price reduces CS. The people sitting in the seats also change. At \$100 there was potential misallocation: some people with a higher MWTP did not get a ticket, which reduced CS conditional on price. With the higher price only 1,000 people demand tickets, and these are the 1,000 highest-MWTP buyers. Misallocation is eliminated, which either raises the sum of MWTP or leaves it the same.

Since the sum of MWTP rises (or stays the same) and P rises, the net impact on Consumer Surplus is uncertain.